COMP 9200: Applied Research Methods
Assignment 3

This assignment has two parts:

1) *an AI-assisted analysis of a public dataset with a comparison to your own analysis in an IEEE-formatted paper (to be done in pairs)*

2) *a first draft of the methodology section of your research proposal (to be done individually)*

Submit your answers to the Learning Hub assignment 3 folder.

1. Choose a public dataset.
   a) Analyze it using a generative AI tool (e.g., Copilot, ChatGPT, Claude) with full prompt/output disclosure.
   b) Analyze it using your own analysis pipeline (Python/R/stats tool).
   c) Compare findings, rigor, transparency, bias, and reproducibility.

   Paper Requirements:
   - 4-6 pages, IEEE conference paper format in LaTeX
   - Sections: Title, Abstract, Keywords, Introduction, Related Work (brief), Methodology (dataset, preprocessing, AI tool setup, your pipeline), Results (AI vs. You), Discussion (interpretation, validity, ethics, reproducibility and AI Transparency), Conclusion, References.

   Example datasets:
   - UCI Machine Learning Repository (e.g., Human Activity Recognition, Eye Tracking)
   - OpenML classification/regression datasets (tabular)
   - Kaggle: user behavior analytics, app usage patterns, app store reviews
   - Public UX/SUS datasets or institutionally approved telemetry (if license permits)
   - GitHub Issues/PR metadata for developer experience (non-PII aggregates)

2. Submit the first draft of the methodology section of your proposal, following the guidelines in the textbook and in class. Ensure that you detail your data collection process and your analysis methods, including how you will ensure validity and reliability of your instruments.

Use the LaTeX templates provided.

Remember to include your AI use statement.